

Causes Of RELAY FAILURE In Switching Power Supplies And The Remedies

In a switching power supply, sometimes the relay contacts fail due to a short-circuit, but the product continues to work. In the long run, this damages the device, especially because it is difficult to find the failure. Relays must be modeled correctly, and contacts must be chosen wisely to prevent such instances

CONTENT TEAM,
MORNSUN
and AARYAA
PADHYEGURJAR

All devices require a power source to work efficiently. Power supplies are devices that convert electric current from a source into the voltage required to power any appliance. A power supply can be of two types—linear or switching. For the scope of this article, we will be focusing on switching power supply and analysing certain failures that are associated with it.

Despite being reliable, cost-effective, and ideal for low power applications, linear power supply has its disadvantages. Switching power supply design was developed to overcome these disadvantages and is comparatively a recent methodology. It uses switching devices that are turned on and off at high frequencies. This is why an electronic switch is one of the most fundamental components of a switching power supply circuit. The two most commonly used switching devices are the bipolar NPN transistor and the N-channel MOSFET. Furthermore, while the switching device is in its non-conduction mode, the switching power supply uses components such as inductors and capacitors to supply power.

Performance indices for switching power supply

Power factor. The power factor of an electri-

cal or electronic device is the ratio of the power it draws from the main supply to the power it actually consumes. A gadget with a power factor of 1.0 consumes all of the energy it draws. It is possible to achieve a unity power factor, but many devices do not.

In the case of electronic devices, such as PCs and chargers that are driven by switching power sources, the power factor may be low because the device pulls current in a non-sinusoidal waveform. This is commonly referred to as the 'distortion' power factor, and it is very important to rectify it. If that is not done, the distorted power factor can cause unwanted harmonics.

Power factor correction. In a perfect system, 100% of the power drawn from the AC mains is put to good use. However, such a scenario is too good to be true. Due to the low power factor, some of the energy from the AC source is not utilised and is lost. This increases cost and contributes to global warming too. Hence, power factor correction (PFC) is done to increase the power factor.

PFC can be done in two ways:

First, passive PFC, which improves PF by utilising passive filters to filter out harmonics. This is commonly used in low-power applications. However, it is insufficient in high-power applications.

Second, active PFC, where a switching converter is used to modify the distorted wave into

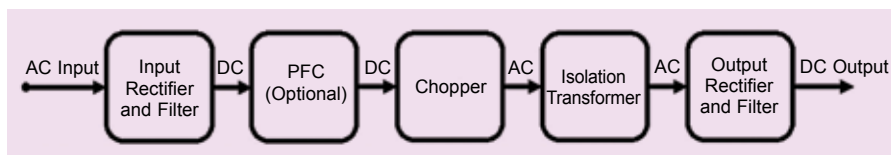


Fig. 1: Block diagram of a switched-mode AC/DC power supply (Credit: www.monolithicpower.com)